AUTHBC: Feasibility Boundaries for Authenticated UAV Telemetry — An Exclusion Bound and a Capacity Envelope

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***Abstract*—Unmanned Aerial Vehicles (UAVs) that broadcast signed, hash-chained telemetry spend more bytes on authentication than on data: a 64 B signature and a 32 B chain link accompany a 24.0 B record. This paper presents AUTHBC, a frame format, sender and receiver in which one signature and one chain link cover b records while every frame decodes and verifies alone. Batch size, encoding and signature scheme are selected jointly under four constraints: freshness, verifiability, channel capacity and receiver CPU. The evaluation combines emitted frames, ns-3 simulation and metered sender energy. The results demonstrate two feasibility boundaries. First, eight of twelve EU863-870 LoRaWAN data rates cannot carry one self-verifying frame (five cannot hold the signature in any format, three our frame). Second, at 50 records/s and a 100 ms deadline the design sends 43.25 B per record against 145.77 B for per-record signing, and one simulated 802.11a collision domain delivers 95% of frames among 124 nodes instead of 35. Batching gains nothing unless record rate times deadline exceeds two. Capacities are simulated, including a spread, moving swarm (gain 3.46–3.63-fold); contention was measured on two radios.**

***Index Terms*—FANET, UAV telemetry, broadcast authentication, digital signatures, hash chain, batching, IEEE 802.11 DCF, LoRaWAN, ns-3**

I. Introduction

Flying Ad-hoc Networks (FANETs) enable Unmanned Aerial Vehicle (UAV) swarms to coordinate without centralized infrastructure, and each vehicle in a swarm reports its position, velocity and status to its neighbors many times per second. When these reports must serve as evidence afterwards — of where each vehicle was, and of what each vehicle claimed — every report must be attributable to its sender and the sequence must be tamper-evident. The standard construction signs every record and links it to its predecessor by a hash. However, this construction introduces a critical challenge once the payload is compact: the protection becomes larger than the data it protects. Our earlier comparison of encoding formats showed that canonical Concise Binary Object Representation (CBOR) reduces an 18-field telemetry message from 358 B in JavaScript Object Notation (JSON) to 130 B [1]. The compactly encoded position, velocity and status record of this work occupies approximately 24.0 B, whereas an Ed25519 or Elliptic Curve Digital Signature Algorithm (ECDSA) signature occupies 64 B and a SHA-256 chain link 32 B. On a shared radio channel these bytes translate directly into airtime, and airtime determines how many neighbors can be heard.

Current practice authenticates each message individually. MAVLink 2 appends a 13 B symmetric tag to each message [2], and vehicular standards attach an ECDSA signature, and periodically a certificate, to each one [3]. Existing stream-signing schemes [4], [5], [6] amortize the cost of signing across packets without reducing the number of frames that a sender places on the channel. Conversely, aggregate-signature schemes [7] reduce the work of the verifier without reducing what each signer transmits. Letting several records share one frame, one signature and one chain link is an established alternative, and its price is known in kind: records wait for the frame to fill, and a lost frame loses all of them. What we did not find is the accounting: what a frame must carry so that a receiver can verify it without any other frame, on which links such a frame does not fit at any batch size, which constraint caps the batch where it fits, and what the saved bytes buy on a contended channel in nodes and not in percent. This gap leaves system designers without data-driven guidance for selecting batch size, encoding and signature scheme jointly.

This paper presents AUTHBC — authenticated telemetry for a blockchain-style, hash-chained ledger — as a frame format, a sender and a receiver. The system was implemented, measured, and searched over its parameters. The main contri-

butions of this work are:

- A capacity rule with its mechanism. Capacity is obtained by simulating each configuration at its own frame size and rate, with 30 seeds and a bootstrap interval, and not by applying one load ceiling to all. A rule with one fitted constant, the capacity envelope of the title, reproduces the simulated capacities. The access rule accounts for the constant and, with one loss mechanism that earlier analyses of broadcast do not contain, yields the capacities with nothing fitted. Every prediction made from either was written down before its runs, other batch sizes and unequal received power among them (Sec. VIII-D).
- A ladder that separates causes. Sharing a frame, sharing a signature and delta coding are separate steps; bytes, capacity and receiver CPU are reported for each (Sec. VIII), and sender energy with and without batching. Most of the capacity is gained by sending fewer frames, the effect that frame aggregation exists for.
- An exclusion bound with its scope. On LoRaWAN, eight of twelve EU863-870 data rates cannot carry one signed, chained frame: five because the signature alone does not fit, and three because of the header of this frame. The five conditions under which the bound holds are stated, together with what each relaxation buys (Sec. VII).
- A frame as one object, and the constraint that binds. The design and its baseline are frames that were emitted, decoded and verified, and their sizes are measured lengths (Sec. IV); the lean format is 42% smaller than the format we built first. Freshness, not the Maximum Transmission Unit (MTU), caps the batch at four records. A delta-coded frame that depends on its predecessor misses the verifiability target at its own loss budget, so every frame carries one self-contained record. Counting cryptography alone, one CPU core serves a neighborhood of 296 nodes with Ed25519 and of 10 with BLS; the interpreted receiver we built serves 57 on one core, and its compiled version 289, so that the channel, not the processor, binds.
- A formal model and two defenses. Origin, integrity, replay protection, equivocation evidence, gap detection and continuity are proven in Tamarin, and two gaps of the threat model are closed: a GNSS-time window refuses delayed frames, and an 8 B group tag refuses the frames of outsiders before they cost a verification (Sec. III).

In summary, batching and header design decide what can run, and among standard 64 B elliptic-curve signatures the choice of scheme does not. AUTHBC introduces no new cryptographic primitive. Each result states whether it is measured, simulated or only modeled, and Section XI collects the limitations.

II. Background and Related Work

Our work intersects per-message authentication, stream signing, aggregate and batch verification, hash-linked records, compact encoding for small payloads, and broadcast channel models.

Per-Message Authentication: MAVLink 2 signing appends a link identifier, a 48-bit timestamp and a 48-bit truncated SHA-256 tag computed with a shared secret [2]. This mechanism authenticates a *link*: any holder of the key can produce any message, so a record cannot be attributed to one vehicle afterwards. Rotta and Mykytyn likewise attach a keyed hash (HMAC) and a timestamp to each broadcast telemetry message of a swarm [8]. Vehicular broadcast employs public-key signatures: ECDSA over each message, with the certificate of the signer in some messages and an eight-octet digest of it in the others [3], [9]. We adopt this trust model, and its per-message cost is the baseline of our evaluation.

Stream Signing: To amortize one signature over many packets, Gennaro and Rohatgi sign the first packet and chain the remaining packets by hashes [4]; one lost packet breaks the chain. Wong and Lam sign a hash tree over a block and send in every packet the hashes needed to reach the root [5]: each packet verifies alone, and the signing cost is amortized but the bytes are not. EMSS tolerates loss by carrying several hashes per packet and a periodic signature packet, and TESLA replaces the signature by a Message Authentication Code (MAC) whose key is disclosed later [6], [10]; TBRD applies TESLA to UAS Remote ID and reports about half the overhead of signatures [11]. Erasure codes spread the hashes and the signature of a block over its packets so that the block verifies when enough of them arrive, for 10 B per packet in blocks of sixteen at 5% loss and a latency of one block [12]. However, these schemes spread *one authenticator across packets*, so packets depend on each other or on a later one. Our design places *several records in one packet*, which a small telemetry record allows: each frame stays self-verifying and the cost moves to latency at the sender. TESLA delays verification at the receiver instead, and a disclosed MAC key provides no non-repudiation, which a ledger requires. Section IX places these schemes at our operating point.

Aggregate and Batch Verification: BLS signatures aggregate [13], [14], and certificateless aggregate signature (CLAS) schemes for vehicular networks build on such constructions [7]. Zhang *et al.* verify many signatures at once at a roadside unit [15], and Rajasekaran *et al.* batch-authenticate a group of drones in a pairing-based scheme, reporting a communication cost that grows linearly with the group [16]. All of these reduce the *work* of the verifier, and an aggregator may forward one short aggregate in place of many signatures. However, each message, as its signer sends it, still carries its own signature material (Sec. IX).

Hash-Linked Records: Linking each record to the hash of its predecessor to make a sequence tamper-evident is due to Haber and Stornetta [17]; blockchain applications on UAVs are surveyed in [18]. We take the chain as given and quantify what it costs on air.

Compact Encoding and Small Payloads: That a signature leaves little room in a small frame is established: Gündoğan *et al.* find 9 B of application data left beside an Ed25519 signature in an IEEE 802.15.4 frame [19], and post-quantum signatures do not fit one 5G broadcast block [20]. For Remote ID,

a 64-octet signature is paged over 25-octet Bluetooth frames with a parity page [21]: the spreading and recovery that our scope excludes. Automatic Dependent Surveillance–Broadcast (ADS-B) is the extreme case: its 112-bit messages leave 56 bits for data, and proposals for public-key authentication distribute one signature over several messages that receivers buffer [22]. Compact encodings [23] and delta coding of time series [24] are established, and our earlier evaluation found canonical CBOR byte-identical across 10,000 trials [1]; the frame of this paper is a canonical CBOR map. A lossy link limits how far a delta chain may run (Sec. VIII-B).

Channel Models: Broadcast frames are not acknowledged and their contention window never doubles, so the saturation model of Bianchi for the IEEE 802.11 Distributed Coordination Function (DCF) [25] does not apply as it stands; we employ the broadcast model of Ma and Chen [26], [27] and validate it against ns-3 [28]. For LoRa, Haxhibeqiri *et al.* model single-gateway scalability [29] and Zirak *et al.* report air-to-air delivery against range on hardware [30]. These models describe the channel alone; our evaluation applies them to each authentication configuration at its own frame size and frame rate.

III. System and Threat Model

Fig. 1 gives an overview. Each UAV codes, chains and signs its records into frames and broadcasts them on a shared channel; each receiver decodes, verifies and stores the frames of its neighbors; and five constraints fix the design parameters (Sec. V).

A. Network and Traffic

N UAVs share one channel and all hear one another: one collision domain, which we call a neighborhood of N nodes. Each UAV produces Λ telemetry records per second and broadcasts them; there are no acknowledgments and no retransmissions. The primary operating point is $\Lambda = 50$ records/s with a freshness bound $D_{\max} = 100\,\text{ms}$: the rate is the companion-computer stream rate of PX4 [31], and the pair satisfies the direct UAV-to-UAV broadcast requirements of 3GPP [32]. A relaxed point of 20 records/s and 250 ms is reported for comparison.

B. Telemetry Record

Each record carries nine fields, all fixed-point integers:

- `ts` – timestamp in milliseconds
- `lat`, `lon` – latitude and longitude in degrees$\times 10^7$
- `alt` – altitude in centimeters
- `vel_x`, `vel_y`, `vel_z` – velocity components in cm/s
- `battery` – battery level in percent
- `mode` – flight mode (enumeration)

In the ledger, each record additionally carries the identifier of its sender, its sequence number and the hash of its predecessor.

C. Ledger

Each UAV maintains an append-only chain of its own records, $prev_hash_i = \text{SHA-256}(rec_{i-1})$, and stores the verified records it hears from others. No consensus layer is included; the subject of this paper is authenticated dissemination. A ledger is audited from the uploads alone: one pass over the frames of a mission, with the public keys of the senders, finds every equivocation and every frame that does not verify; a planted equivocation and a planted forgery were both found, with no false finding (Table I). Keeping the frames costs 42.9 B per record.

D. Adversary and Security Goals

Adversary: The adversary controls the radio: it reads, injects, replays, modifies, delays and drops frames, and may operate many radios. It holds no private key of an honest UAV.

What Is Protected: (i) *Origin and integrity:* a stored record was produced by the UAV it names and was not altered, by a public-key signature checked before anything is stored. (ii) *Replay:* sequence numbers are strictly increasing per sender; a repeated frame is refused. (iii) *Equivocation:* two validly signed records for one sequence number are detected, and the two frames that carried them are kept. The signature covers a frame, so a record is evidence only with its frame; a third party checks the pair with the sender's public key alone. (iv) *Continuity:* omission or reordering within what a receiver holds is detected through the chain. (v) *Bounded-delay freshness:* while GNSS time is held, a receiver refuses a frame stamped more than Δ before its own clock reads (Sec. VIII-C).

What Is Not Protected: Telemetry is sent in clear. Jamming is not resisted: loss appears as the robustness measure V below, not as a security property. Flooding is resisted only from outside the swarm: an 8 B tag under a key that all members hold refuses the frame of an outsider before it is decoded, but a member holding that key can still make a receiver verify forged frames (Sec. VIII-D). A frame delayed by less than Δ is accepted, authentic and in order, and so is a later one once GNSS time has been lost for longer than the window allows. A UAV with a valid key can sign false data; only inconsistent falsehood is detectable. Key distribution, revocation and admission are assumed: receivers hold the public keys of the senders, and the bytes of delivering certificates are charged separately (Sec. VIII-E). Senders are not anonymous.

Formal Analysis: Properties (i)–(iv) and the detection of gaps are proven in the Tamarin prover [33] for unboundedly many frames, and (v) is shown to hold with the window and to fail without it (Table I). The proofs use three models, since one model with both the chain and the clock did not terminate; canonical encoding is assumed injective.

E. Verifiability Metric

V is the probability that a record a UAV generates is received, decoded and verified by a given neighbor. We require $V \geq 1 - \epsilon$ and take $\epsilon = 0.05$ throughout.

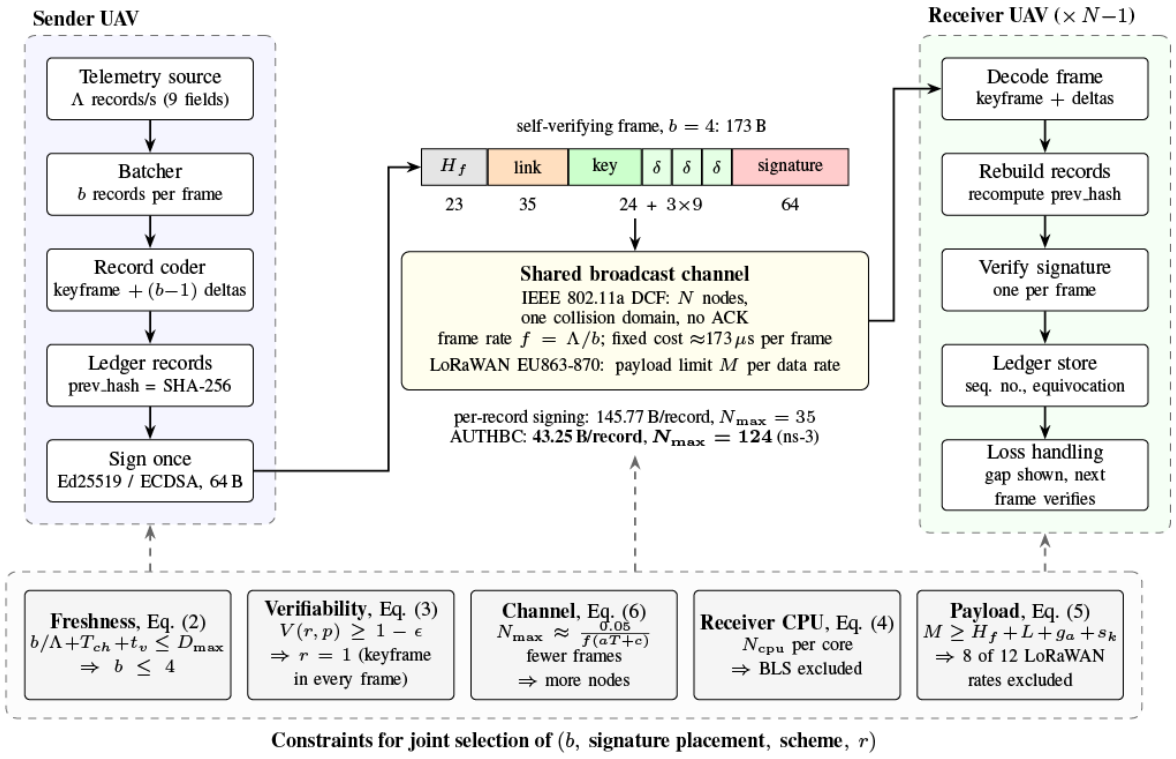


Fig. 1. System overview: sender pipeline, the four-record frame with its field sizes (bytes), shared channel, receiver pipeline, and the five constraints that select the design.

IV. The AUTHBC Frame

A. Frame Structure

A frame is one canonical CBOR map [23] with integer keys (Table II, Fig. 1); Appendix A provides the complete layout. The fields `src` and `base_seq` name the sender and the sequence number of the first record; `link` is the hash of the record before it; `recs` is the record stream and `auth` one signature. A record in the stream consists of nine variable-length integers. The first record of a frame is a *keyframe*, carrying absolute values (24.0 B on average); each later one is a *delta* from the record before it (9.0 B, the floor of the format, at the 20 and 50 ms spacings of our two operating points). Sender and sequence number are not repeated in the records.

B. Signed Content and Verification

The signature covers not the record stream but the canonical ledger records, each with its own `prev_hash`. The receiver rebuilds them — the first `prev_hash` is `link`, and each later one is the hash of the record just rebuilt — and only then checks the signature. A frame that cannot be decoded therefore cannot be verified, and nothing unverified is stored. Each verified record is offered to the ledger store, which refuses a sequence number that is not newer, keeps an equivocating pair as evidence, and flags a link that contradicts a record it holds. The state against which a later frame would be decoded advances only after a frame has been verified and stored, so a replayed frame cannot desynchronize a stream. Flipping any single bit of a frame makes verification fail (all 1 392 positions of a four-record frame were tried).

C. Behavior Under Frame Loss

When a frame is lost, the next one still decodes and verifies, because it starts with a keyframe. Its `link` then names a record that this receiver does not hold: the gap shows in the sequence numbers, and the link is stored unchecked, to be checked by any party that holds the missing record. A late copy of the lost frame is refused, since sequence numbers must increase; the gap is not repaired.

D. One Chain Link per Frame

The ledger still chains every record. Only the first link of a frame travels; the receiver recomputes the others. Inside a frame, the order and content of records therefore rest on the frame signature alone, whereas in the first format below every record also carries its own hash on air. A frame is signed and accepted as a unit, so those hashes were redundant and a receiver loses no check; however, one mechanism remains where there were two.

E. The First Format

For comparison we retain the format we built first: text keys, and in every record its own chain link, sender identifier and sequence number. Its header is 44 B, of which 29 B are key names. That format is implemented and frozen. Its sizes

TABLE I
SECURITY PROPERTIES AND THREATS, THE MECHANISM FOR EACH, AND THE EVIDENCE: PROOFS IN TAMARIN, SIMULATED CLOCKS, AND FLOODS AGAINST ONE RASPBERRY PI 4 RECEIVER CORE.

Property or threat	Mechanism	Evidence	Result
(i) Origin and integrity	signature over the frame	Tamarin, model without chain and clock	proven
(ii) Replay	increasing sequence numbers	same model	proven
(i) and (ii) together	both	same model	injective agreement proven
(iii) Equivocation	both frames kept	same model	proven: it implies a revealed key
Gap in a sender's frames	sequence numbers	same model	detection proven
(iv) Continuity	chain link per frame	Tamarin, model with the chain	proven
(v) Delayed frame	GNSS-time window Δ	Tamarin, model with the clock simulated clocks, 72 000 honest frames per setting	accepted without the window, refused with it with $\Delta = 150$ ms no honest frame refused at clock errors up to 10 ms; every frame held 250 ms or more refused; a 250 ms window outlasts 75 min without GNSS
Flooding by outsiders	8 B group tag	UDP flood, 41 senders on one core	$V = 0.984$ at 3 000 forged frames/s (Fig. 10); the tag costs 3.1% of capacity
Flooding by a member	none	same	$V < 0.95$ from 129 forged frames/s
Ledger audit	signed frames uploaded	three UAVs, one flight hour, 391 495 frames	both planted faults found, no false finding, in 6.4 min of one core

TABLE II
BYTE BUDGET OF A FRAME BY FIELD IN THE FIRST AND THE LEAN FORMAT.

Field	Content	First	Lean
(map)	CBOR map header	1	1
`v`, `t`	version; frame type	6	4
`src`	sender identifier, 16 bit	7 (5)	4 (2)
`base_seq`	sequence no. of first record	14 (10)	6 (2)
`n`	records in the frame	3	2
`recs`	key, length of records	6	3
`auth`	key, length of signature	7	3
header H_f †		**44** (38)	**23** (17)
chain link		32 per record	35 per frame
signature g_a (Ed25519, ECDSA)		64	64
keyframe record s_k		59.9*	24.0
delta record s_δ, 50 ms apart		44.0*	9.0

† five-digit sender identifier, one hour into a 50 Hz flight; in parentheses: sender 0, first frame of a flight.
* includes the 32 B chain link of that record.

in this paper are sums of separately measured parts, and its delta-coded frames were never emitted; such sizes are marked wherever they appear.

F. Design Parameters

A configuration is defined by four parameters:

- *Batch size* b – the number of records in a frame.
- *Signature placement* – a signature covers each record (*inline*; `auth` is then an array of b signatures) or the frame (*batch*).
- *Signature scheme* – Ed25519 [34] and ECDSA P-256 [35] at 64 B, or BLS12-381 [13] at 96 B in the library used.
- *Reference interval* r – one frame in r starts with a keyframe, and the others start with a delta against the previous frame.

V. CONSTRAINTS

Bytes: With header H_f, link field L, signature g_a, keyframe s_k and delta s_δ, a self-contained batch frame costs, per record,

$$B(b) = \frac{H_f + L + g_a + s_k + (b-1)\, s_\delta}{b}. \tag{1}$$

$B(1)$ is our baseline: one record, one signature, one frame. Signing each record of a shared frame replaces g_a by $b\, g_a$. Equation (1) is exact to within a byte of CBOR length prefix; for the lean design and its baseline we report the lengths of emitted frames.

Freshness: The oldest record in a frame waits for the frame to fill. Adding the channel-access and airtime delay T_{ch} and the verification time t_v,

$$D(b) = \frac{b}{\Lambda} + T_{ch} + t_v \;\le\; D_{\max}. \tag{2}$$

Fill time dominates. At both operating points $\Lambda D_{\max} = 5$: five records fill in exactly $D_{\max}$ and leave nothing for the channel, so $b \le 4$ whatever the encoding or the scheme. Equation (2) counts the whole filling period b/Λ. A record may describe a state up to one sampling interval old; and with b/Λ the bound is on the *peak age* of information [36] at a receiver that loses no frame: the delay of the newest record of a frame plus the interval to the next frame. Counting only the wait $(b-1)/\Lambda$ would admit $b = 5$ and 36.4 B per record. The 1500 B MTU would admit 31 records in the first format and 151 in the lean one.

Verifiability: If one frame in r is self-contained, frame j of a group verifies only if it and the $j-1$ frames before it arrive. With independent frame loss p,

$$V(r,p) = \frac{1}{r}\sum_{j=1}^{r}(1-p)^j, \tag{3}$$

which equals $1-p$ at $r = 1$ and is smaller for every $r > 1$. A target $V \ge 1-\epsilon$ with $\epsilon \le p$ therefore admits $r = 1$ only. The

TABLE III
EXPERIMENTAL PLATFORM SPECIFICATIONS.

Parameter	Simulation Host	Embedded Board
Platform	Intel Core i5-14400F	Raspberry Pi 4 Model B Rev 1.4
CPU	x86-64	Cortex-A72, 4 cores, 1.8 GHz
OS	Linux (WSL2)	Debian 12 (bookworm)
libc	glibc 2.39	glibc 2.36
Python	3.12.3	3.12.13
Role	simulation, analysis	timing, energy, radio

same argument forbids splitting a signed unit over n frames, which verifies with probability $(1-p)^n$.

Channel: For a configuration — a frame size and a frame rate per node — $N_{\max}$ is the largest neighborhood in which the mean delivered fraction of frames is at least $1-\epsilon$. It is obtained by simulation (Sec. VI). As a conservative companion we report N_{sat}, the largest N whose offered frame rate does not exceed the saturation throughput of the broadcast DCF model [27].

Receiver CPU: A receiver verifies the frames of every neighbor. With k signatures per frame, verification time t_v and hash time t_h, one neighbor costs $(\Lambda/b)\,k\,t_v + \Lambda\,t_h$ seconds of CPU per second, and one core serves

$$N_{\mathrm{cpu}} = 1 + \left\lfloor \frac{1}{(\Lambda/b)\,k\,t_v + \Lambda\,t_h} \right\rfloor \tag{4}$$

nodes, itself included.

Payload: A frame that verifies alone must fit the maximum payload M of the link:

$$M \;\geq\; H_f + L + g_a + s_k. \tag{5}$$

VI. Experimental Setup and Methodology

A. Implementation Configuration

Formats, sender, receiver and ledger store form one Python code base with four record encodings (JSON, CBOR, MessagePack, delta) and three signature schemes behind common interfaces. All components use standard libraries in their default configurations, without hand-tuned optimization:

- cbor2 v5.8.0: canonical encoding [23] of frames and of the ledger records that a signature covers.
- cryptography v45.0.7: Ed25519 and ECDSA P-256 signatures; SHA-256 is that of the Python standard library.
- blspy v2.0.3: BLS12-381 signatures.
- ns-3.48 [28]: channel simulation for IEEE 802.11a and 802.11p, and with its LoRaWAN module for the low-rate links.
- Rust with ed25519-dalek: a compiled receiver and sender, byte-identical to the Python codec on 10 000 frames.

Each signature scheme is checked against published test vectors (for Ed25519, those of [37]). Table III lists the two platforms; the board runs with the `performance` frequency governor.

B. Telemetry Workload

Records are produced by a seeded random-walk generator: 30 seeds of 1000 records at 50 ms spacing, that of the relaxed point. Header fields are sized for a five-digit sender identifier one hour into a 50 Hz flight. Section VIII-G checks the generator against flight logs.

C. Measurement Methodology

Timing Instrumentation: Signing, verification, hashing and encoding are timed on the Raspberry Pi 4 and reported as medians: Ed25519 sign/verify 88/259 μs, ECDSA 136/327 μs, BLS 2.73/8.61 ms, and SHA-256 2.75 μs.

Energy Measurement: Sender energy is metered with an INA219 on the supply rail, five repetitions per configuration, idle power subtracted: the first format on that board, the lean format, interpreted and compiled, on a second board and sensor of the same types.

Channel Simulation: The channel is simulated in ns-3.48 [28]: IEEE 802.11a at 6 Mb/s, ad hoc mode, broadcast frames, all nodes in range at equal received power, so there is no capture and no hidden terminal. Each node sends one frame every b/Λ seconds, at an instant drawn uniformly within each period, so the relative phases of the nodes change from frame to frame. A strictly periodic source is not used, because it freezes every phase for a whole run: a run then samples one phase configuration, in which a few nodes may collide in every period while the others lose nothing. The two sources deliver the same on average — over 14 points and 1 260 runs the means differ by 0.0004 ± 0.0009 — but the strictly periodic one varies 6–23 times more from run to run (Sec. XI).

Statistical Treatment: A run lasts 20 s and each point comprises 30 runs. $N_{\max}$ is the largest simulated N whose mean delivered fraction is at least 0.95, every smaller simulated N also passing. Its interval is the 2.5–97.5 percentile of that quantity over 10 000 bootstrap resamples of the seeds. An interval of a mean over seeds is a Student-t interval, and an interval of an interpolated crossing an expanded percentile bootstrap interval [38]: with a known truth and 30 seeds, the plain percentile interval covered 92–94% of the time and these 94–95%. The saturation models were validated separately against saturated ns-3 runs (Sec. VIII-H).

The main evaluation uses one collision domain of static nodes; Section VIII-D repeats it for a swarm spread over a wider area, with hidden stations, and moving. Capacities are obtained by simulation, and contention was measured on hardware between two radios only (Sec. XI).

VII. Low-Rate Links: Where No Batch Helps

LoRaWAN limits the payload of a frame by data rate, and its EU863-870 plan defines twelve data rates [39]: seven LoRa (DR0–DR6), one FSK (DR7) and four LR-FHSS (DR8–DR11). By Eq. (5), a frame that verifies alone requires a header, a chain link, a signature and one record that decodes alone. The smallest such frame that the lean format can emit at all — sender 0, sequence 0, every field zero — is

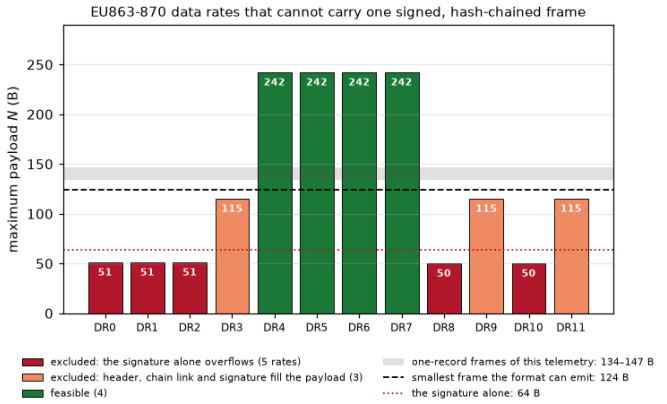


Fig. 2. Payload limits of the twelve EU863-870 data rates against the smallest lean frame and the frames of this telemetry (64 B signature).

TABLE IV
FIT OF ONE SELF-CONTAINED LEAN FRAME WITHIN THE EU863-870 PAYLOAD LIMITS, BY AUTHENTICATOR.

	Floor† (B)	**Range**‡ (B)	**50/51 B** 5 rates	**115 B** 3 rates	**222/242 B** 4 rates
64 B signature, chain link	124	134–147	no	no	yes
… link not sent on air	89	99–112	no	yes	yes
48 B BLS signature	108	118–131	no	no*	yes
13 B symmetric tag	72	82–95	no	yes	yes

† every field zero. ‡ frames of this telemetry.
* a frame of zeros would fit; no frame of this telemetry does.

$16 + 35 + 64 + 9 = 124$ B; this frame was emitted, signed and verified. Header, link and signature alone occupy 115 B.

Fig. 2 compares that floor with the payload limits, revealing three key findings:

1) *Signature-Bound Rates:* Five data rates (DR0–DR2, DR8, DR10) allow 50 or 51 B: the signature alone does not fit.
2) *Format-Bound Rates:* Three data rates (DR3, DR9, DR11) allow 115 B: header, link and signature fill the payload and no record fits.
3) *Feasible Rates:* Four data rates (DR4–DR7) allow 222 or 242 B and can carry a frame.

Eight of twelve data rates cannot carry one signed, chained frame, in either format and under either of the payload tables of the standard. This exclusion bound does not depend on the telemetry, since each rate is excluded against the floor of the format; three of the eight depend on the format (condition v).

Scope: The result holds under five conditions. (i) The limits are the regional parameters of LoRaWAN; the LoRa PHY itself carries 255 B [40] and is not excluded. (ii) Every frame carries a public-key signature. (iii) There is no loss recovery. (iv) $\epsilon \leq p$, which by Eq. (3) forbids spreading a signature over several frames; at $p = 0.01$ and $\epsilon = 0.05$ a signed unit may span five frames and the exclusion no longer holds as stated. (v) The frame is the format of Section IV. The five rates of 50 and 51 B are excluded by the signature alone, whatever the format; at 115 B a signature and a 32 B hash leave 19 B, which the header and framing of this format fill exactly. A hand-packed layout would leave part of them for a record, less than the 24.0 B that our keyframe averages.

Effect of Each Relaxation: Table IV quantifies what each relaxation buys. Taking the chain link off the air frees the three 115 B rates: they are excluded by the link, not by the signature. The smallest standardized signature, 48 B BLS12-381 [41], fits a 51 B payload and leaves 3 B; at 115 B a frame of zeros would fit (108 B) but the smallest frame of our telemetry is 118 B, a three-byte miss that we do not count as an exclusion. A 13 B symmetric tag, as in MAVLink 2, fits the 115 B rates, but at 50 or 51 B header and link already take 51–58 B: a hash-chained frame does not fit there under any authenticator.

What the Feasible Rates Carry: At DR5 under the 1% duty cycle a 242 B frame holds 8 records and sustains 0.20 records/s per node (5 records and 0.13 in the first format). The batch must be sized at the record spacing it implies: a delta that spans 5.5 s costs 13.6 B, not the 9.0 B it costs at 50 ms. The ns-3 LoRaWAN module carries at most 222 B, where a frame holds six records. Simulated with a 218 B payload every 36.4 s (0.165 records/s per node), on one channel with one demodulator and no capture, the largest neighborhood with mean $V \geq 0.95$ is three nodes (30 seeds, 95% interval $[2, 3]$; one if 95% of runs must each reach it); the closed-form model of Haxhibeqiri *et al.* [29], fitted to hardware interference measurements and evaluated at the same operating point, gives four. Such a neighborhood carries about 0.5 records/s in all, where the 802.11 design of Section VIII carries some 6 200. On these links the ledger therefore records provenance every few seconds; it is not the 802.11 design at a smaller scale.

VIII. RESULTS ON 802.11

Terminology: Bytes per record is the on-air length of a frame divided by the number of records it carries, $B(b)$ of Eq. (1). $N_{\max}$ is the largest neighborhood with a mean delivered fraction of at least 0.95 in ns-3, and N_{sat} is the saturation bound of the broadcast model. The capacity ratio is the $N_{\max}$ of a design divided by that of its baseline; it depends on the format and is therefore always quoted with the format.

A. The Design Ladder

Table V presents the design ladder, which separates the steps that a per-record baseline confounds, revealing three key findings:

1) *Sharing the Frame:* When four records share a frame but each keeps its signature (rows 2 and 6), the lean format falls from 145.77 B per record to 93.00 B: the header and the chain link are paid once per four records, and the three records that follow the first are coded as deltas.
2) *Sharing the Signature:* One signature per frame yields **43.25 B, 70.3% below the baseline**; row 7 is the same frame without delta coding, 54.50 B.
3) *Position of the Chain Link:* In the first format the same steps are sharing the frame (174.25 to 141.25 B), sharing the signature (93.25) and delta coding (74.96, 57.0% below its baseline). It ends higher because each of its

TABLE V

DESIGN LADDER AT 50 RECORDS/S AND $D_{\max} = 100$ MS: FRAME SIZE, BYTES PER RECORD, SATURATION BOUND N_{sat}, SIMULATED CAPACITY $N_{\max}$ WITH ITS 95% INTERVAL, AND RECEIVER CPU SHARE.

	Configuration	b	Frame (B)	B/record	With cert., one per 5 frames	With cert., one per 500 ms	N_{sat}	$N_{\max}$	CPU*
First format: text keys, a chain link in every record†									
1	sign every record, one record per frame (baseline)	1	174.3	174.25	213.05	188.41	18	32 [32, 32]	41%
2	sign every record, share the frame	4	565.0	141.25	150.95	149.41	24	56 [56, 56]	72%
3	one signature per frame	4	373.0	93.25	102.95	101.41	30	76 [75, 76]	25%
4	… and delta records	4	299.9	74.96	84.66	83.12	34	88 [88, 89]	29%
Lean format: integer keys, one chain link per frame									
5	sign every record, one record per frame (baseline)	1	145.8	145.77	184.57	159.93	20	35 [35, 35]	45%
6	sign every record, share the frame (delta records)	4	372.0	93.00	102.70	101.16	30	76 [76, 77]	98%
7	one signature per frame, no delta coding†	4	218.0	54.50	64.20	62.66	42	108 [107, 109]	36%
8	one signature per frame, delta records **(the design)**	4	**173.0**	**43.25**	52.95	51.41	52	**124** [124, 125]	42%

† size is a sum of measured parts; rows 5, 6 and 8 are emitted frames. * share of one receiver core spent verifying and hashing at $N_{\max}$.

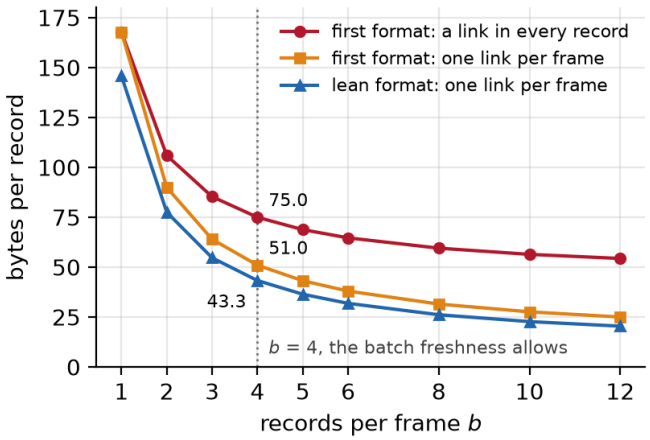


Fig. 3. Bytes per record against batch size, by the position of the chain link (delta records throughout). First-format curves are sums of measured parts; lean frames are emitted.

records carries a 32 B link of its own, which no batch size amortizes.

Fig. 3 quantifies the position of the link. Moving the link of the first format from the record to the frame saves $32\,(1 - 1/b)$ B per record: 24 B at $b = 4$, which gives 51.0 B. The lean header and record coding account for the remaining 7.7 B. With a link in every record, 12 records per frame still cost 54.3 B each, more than one link per frame costs at four. An exhaustive search over placement, batch sizes 1–12 and reference interval returns row 8 as the byte-minimal configuration that meets the freshness, verifiability and MTU constraints, and row 4 for the first format.

Joint Selection: The parameters are selected jointly because they are not independent. One signature per frame saves $g_a(1-1/b)$ per record, and delta coding saves $(1-1/b)$ of the difference between a self-contained record and a delta, since each frame must start with a self-contained one. Both vanish at $b = 1$, where rows 1 and 5 already are the design. Batching is the choice on which the other two gains depend, and the deadline sets it.

TABLE VI

VERIFIABILITY AT FRAME LOSS $p = 0.05$ WITH ONE SELF-CONTAINED FRAME IN r ($b = 4$, LEAN FORMAT; 30 SEEDS, 95% INTERVAL).

r	B/record	V measured	Eq. (3)	$V \geq 0.95$
1	43.25	0.952 [0.947, 0.957]	0.950	yes
2	41.38	0.921 [0.916, 0.927]	0.926	no
4	40.44	0.876 [0.863, 0.890]	0.881	no
16	39.73	0.655 [0.616, 0.693]	0.665	no

Signature Scheme: Among the 64 B schemes the signature scheme cannot move the bytes or the capacities of these rows: Ed25519 and ECDSA produce frames of one size. They differ in verification time (Ed25519: 259 μs vs. ECDSA: 327 μs), which enters Eq. (4) only. BLS adds 32 B per frame and, as shown below, fails on CPU.

B. Loss and the Keyframe Requirement

Delta coding is usually run with a keyframe every K records. With $K = 16$ and $b = 4$ only one frame in four decodes without its predecessor, and the frames that follow a lost one cannot be decoded until the next keyframe. Table VI presents this measurement with the codec in the loop — 7 200 frames per cell through the sender, an independent loss process and the receiver — revealing three key findings:

1) *Dependent Frames Miss the Target:* At the loss rate the system is specified for, $r = 4$ verifies 0.876 of records, not 0.95, and Eq. (3) predicts the measurement at every r. Under bursty loss a dependent design loses less — a Gilbert process with the same mean and bursts of four frames gives 0.929 at $r = 4$ — but it does not reach $1 - p$.
2) *Limited Prize:* A cleaner link would admit a longer interval ($r = 4$ at $p = 0.02$, $r = 8$ at $p = 0.01$), and the most a longer interval can save is 3.5 B per record.
3) *Specification Versus Measurement:* The loss rate here is a specification, not a measurement: at the 2.3×10^{-4} we measured between two radios (Sec. VIII-H) every

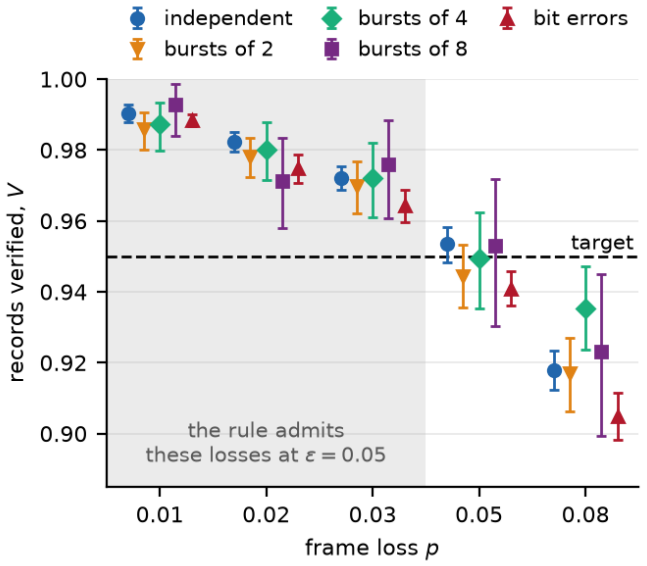


Fig. 4. Records verified by the design ($b = 4$, $r = 1$) against frame loss, five loss processes, 95% intervals; with bit errors, p is one-record frame loss.

interval in the table meets the target, $r = 16$ with $V = 0.998$.

The design is sized for the budget and therefore uses $r = 1$: each frame starts with a keyframe, costs 43.25 B per record, and loses only its own records when it is lost.

Loss in the Simulator: Traced at the capacity crossing, 3 000 neighbor links in ns-3 lose frames independently of one another: bursts average 1.06 frames, against 1.055 for independent loss at the same rate. Eq. (3) is therefore the model of V at the crossing, and the pessimistic one where losses are bursty. Fig. 4 shows the design under five loss processes with the same mean: with $\epsilon = 0.05$ and $p \leq \epsilon - 0.02$ it meets the target, intervals included, under every one of them. At a looser budget the margin can be too small: at $\epsilon = 0.1$ and $p = 0.08$ the design's interval falls below the target under bursts of eight frames and under bit errors, though its mean does not.

Length-Dependent Loss: These values give every frame the same loss probability. At a fixed bit error rate a longer frame is lost more often, and a batch is a longer frame. On a channel that loses 5% of the one-record frames of the baseline, the frame of the design is lost with probability 0.057: 0.943 of its records are expected to verify, below the target, and the measurement, 0.947 [0.941, 0.953], agrees with that expectation. The design reaches 0.95 only while one-record frames are lost at most 0.044 of the time. The capacities of Table V are simulated at equal, strong received power, where frames are lost by collision and not by bit error, and carry that qualification.

C. Freshness

Table VII presents the terms of Eq. (2). The frame fills in 80 ms; channel access and airtime add 0.76 ms on average and 3.49 ms at worst; one Ed25519 verification adds 0.26 ms. The oldest record is verified within 83.8 ms, leaving 16.2 ms of the 100 ms budget.

TABLE VII
DELAY OF THE OLDEST RECORD IN A FRAME ($b = 4$, 50 RECORDS/S); CHANNEL DELAY SIMULATED AT THE HEAVIEST LOAD THAT STILL DELIVERS 95% OF FRAMES.

	Fill (ms)	Channel (ms)	Verify (ms)	Total (ms)	Margin (ms)
Ed25519, mean channel delay	80	0.76	0.26	81.0	19.0
Ed25519, 99th percentile	80	2.40	0.26	82.7	17.3
Ed25519, worst observed	80	3.49	0.26	83.8	16.2
BLS, worst observed	80	3.49	8.61	92.1	7.9

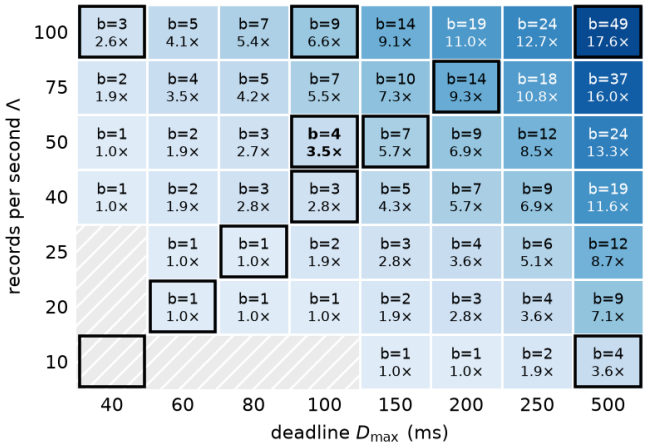


Fig. 5. Design map: the batch b^* the deadline admits and the capacity it buys over per-record signing, by the access-rule model; outlined: simulated in ns-3; hatched: none.

Deadline Compliance: Batching b records requires a record rate above $b/D_{\max}$, here 40 records/s: at 20 records/s with the same 100 ms bound the batch is one record and nothing is gained. At other rates and deadlines Eq. (2) admits other batches; at 15 cells of rate and deadline the simulated capacities lie within 1.05% of the access-rule model, and at 100 records/s and 100 ms, where it admits a batch of 9, the design serves 6.63 times the nodes of per-record signing (Fig. 5).

Age Under Loss: A lost frame adds one frame period, b/Λ, to the age of the next update. At $p = 0.05$ the mean peak age is 87.7 ms, within $D_{\max}$, but its 99th percentile is 169.8 ms: 4.7% of updates exceed the deadline, the share that V also loses.

Delayed Frames: With a window of 150 ms, in 72 000 honest frames per setting no frame was refused at clock errors up to 10 ms, and every frame held back 250 ms or more was refused. In a GNSS outage two clocks part by 144 ms an hour: a fixed window of 250 ms then holds for 75 minutes, and one widened by that drift keeps working, its bound growing at the same rate.

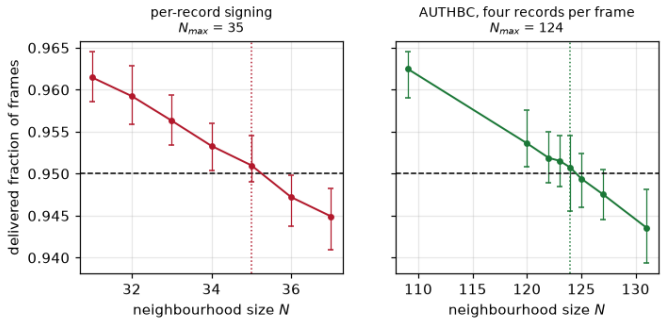


Fig. 6. Delivered fraction of frames against neighborhood size in ns-3: lean baseline (row 5) and design (row 8), mean and range of 30 runs.

TABLE VIII
$N_{\max}$ OF BASELINE AND DESIGN AT BOTH OPERATING POINTS (NS-3, 30 SEEDS, 95% INTERVAL).

	Baseline	Design	Ratio
50 records/s, 100 ms			
first format	32 [32, 32]	88 [88, 89]	2.8×
lean format	35 [35, 35]	124 [124, 125]	3.5×
20 records/s, 250 ms			
first format	78 [77, 78]	219 [215, 221]	2.8×
lean format	85 [85, 86]	306 [306, 308]	3.6×

D. Capacity

Bytes matter because they are airtime. Fig. 6 shows delivery falling as nodes are added, and Table V gives the resulting $N_{\max}$ for every row, revealing three key findings:

1) *Capacity Gain:* The lean baseline holds 35 nodes at 95% delivery and the design 124 (3.5×); in the first format the values are 32 and 88 (2.8×).
2) *Dominant Step:* The larger step is sending a quarter as many frames: row 6, which shares the frame but not the signature, already reaches 76 (2.2× the baseline), and sharing the signature as well adds 1.6×. Each frame pays a preamble, an inter-frame space, a backoff and MAC framing whatever its length — about 173 μs, 47% of what a baseline frame costs the channel — so removing three frames in four is worth more than their bytes.
3) *Stability Across Operating Points:* Table VIII repeats the comparison at the relaxed operating point. The ratio hardly moves — 3.5× and 3.6× in the lean format, 2.8× and 2.8× in the first — and depends on the format, so we quote it with the format.

The dominant step is the principle of frame aggregation in IEEE 802.11n, whose aggregates are addressed to one receiver and exclude broadcast [42]; batching at the application gives a broadcast sender the same effect.

Per-Run Reading: These values hold the mean over runs to the 0.95 target. Requiring that 95% of individual runs reach it gives 34 and 120 (31 and 86 in the first format), at most four nodes fewer.

The Capacity Envelope: At the 5% level, loss follows a line in airtime. With $f = \Lambda/b$ the frames per second a node sends

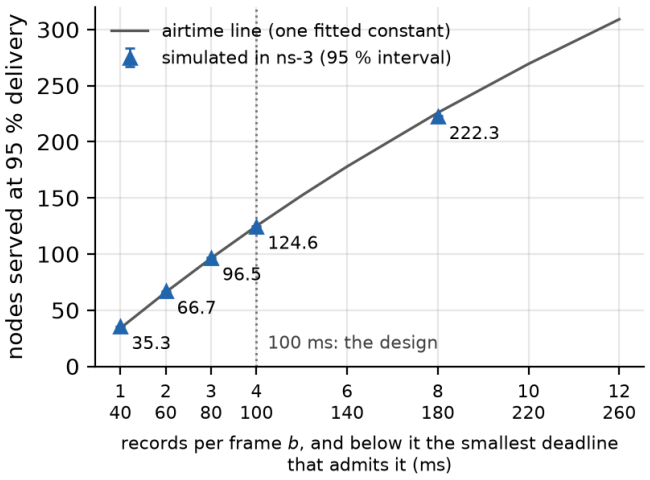


Fig. 7. Nodes the lean design serves at 50 records/s, by batch size and the deadline that admits it: simulated crossings of 0.95 delivery, and the rule.

and T the time one frame holds the medium (its airtime plus DIFS),

$$N_{\max} \approx \frac{0.05}{f\,(a\,T + c)}, \qquad a = 0.071, \quad c = 8\,\mu\text{s}, \tag{6}$$

where c is twice the preamble-detection time of the simulator and a is the one fitted constant. It was fitted to six rows of Table V (1, 2, 4, 5, 6 and 8), which the rule then reproduces within 3.8%. Twelve further configurations were predicted from the rule before they were simulated: seven others of our own, all within 3.4% of it, and the five one-record frames of Table XIV, within 4.2% and all above it. A single utilization ceiling, measured at one point and applied to all, misses the first thirteen (six fitted, seven predicted) by up to 15%, in both directions. Equation (6) is a fit at one loss level and one PHY rate, checked for frames of 94–565 B, 5–50.5 frames/s and 28–306 nodes; the tables quote simulated values, not the rule. All of those configurations hold one record or four. Batches of two, three and eight, which deadlines of 60, 80 and 180 ms would admit, were predicted before they were simulated: delivery crosses 0.95 at 66.7, 96.5 and 222.3 nodes, within 1.6% of the rule and 0.8% of the access-rule model described next. Fig. 7 sets them on the rule: it is what a deadline buys at this record rate, and at 40 ms it buys nothing.

Mechanism of the Crossing: The slope is not a free constant. A frame is lost in this scenario in two ways. In the first, two stations that found the medium busy draw the same backoff slot. With occupancy $\rho \approx NfT$ and a contention window of $W = 16$ slots, counting the stations that wait together gives a loss of about $\rho^2/(W(1-\rho))$, hence $a = \rho/(W(1-\rho))$, which is 0.071 at $\rho = 0.53$, the mean occupancy at which loss reaches 5%. This mechanism is known from analyses of vehicular safety broadcast [43], [44]. In the second, which those analyses do not contain, a station decides to send within the 4 μs in which it cannot yet sense another: that is c. Without the first-order step, the count is $\rho\,[1-(1-1/W)^{\rho/(1-\rho)}]$; with

the detection term and nothing fitted, solving it for 5% gives the 18 crossings within 5.5% (mean 2.3%).

An event simulation of the access rule alone, with nothing fitted and none of the code of the simulator, reproduces all 18 simulated crossings within 1.2%. For crossings that had not been simulated — the contention window doubled, and loss levels of 2% and 10% — it predicted all six within 1.7%. Doubling the window moves the crossing of the baseline only from 35.3 to 39.2 nodes (predicted 39.1): stations that count down for longer wait together more often. The capacities therefore follow from the access rule and one detection time, and they are the capacities of radios that keep that rule (Sec. VIII-H).

Unequal Received Power and Capture: All of this puts every node at one point: equal received power, so a frame that overlaps another is lost everywhere. With the nodes placed uniformly in a disc and power falling as in free space, a receiver keeps the strongest of the frames that start together if it is 4 dB above the rest. The access-rule model with that one rule, read in the source of the simulator, gave 39.5 and 139.7 nodes for baseline and design before the runs. In a disc of radius 15 m, where every station still senses every other by energy, ns-3 gives 39.6 and 140.4; at 100 m, the scale of a swarm, 39.4 and 139.3, at most 0.7% lower. The equal-power capacities of Table V are therefore conservative for unequal power, by 12–13%, and the ratio of design to baseline stays 3.5×. Both discs keep every pair in range: hidden stations, which would lower the capacity, are not in this test.

A Spread, Moving Swarm: Hidden stations appear when a swarm is spread wider than one sensing range. With nodes in discs of radius 100 m to 1 km, log-distance path loss, Nakagami fading and the same capture rule, and in a grouped and a moving layout, the design keeps its advantage: the ratio of the crossings lies between 3.46 and 3.58 in every layout (Fig. 8). At a radius of 1 km, 11% of the nodes able to spoil a frame cannot hear its sender. Counted in neighbors within reach, capacity falls, to 8.7 at 1 km, because a receiver contends with every node in its carrier-sensing range [45]; counted in nodes over the area, it rises to 218. With every node on its own Gauss–Markov path at up to 15 m/s [46], delivery changes by at most 0.0005 and the ratio stays 3.57–3.63; the channel model has no Doppler and no time-correlated fading.

Choosing the Batch at Run Time: A controller that sets b from the occupancy of the channel it measures was simulated on a swarm that grows while it flies. As registered, with a margin of 0.01, it met $V \geq 0.95$ in 75% of one-second windows and failed at 76 nodes; with a margin of 0.02 it met it in 93% and failed at 116, against 99% and 127 with $b = 4$ fixed, while saving 21 ms of peak age (Fig. 9). These ages include the instant at which the simulated source sends each frame, drawn over the next period (Sec. VI), and are on average half a frame period longer than those of Sec. VIII-C. Choosing b at run time buys freshness at the cost of reliability margin; it does not raise capacity.

The nearest published mechanism adapts the contention window to a flocking swarm [47].

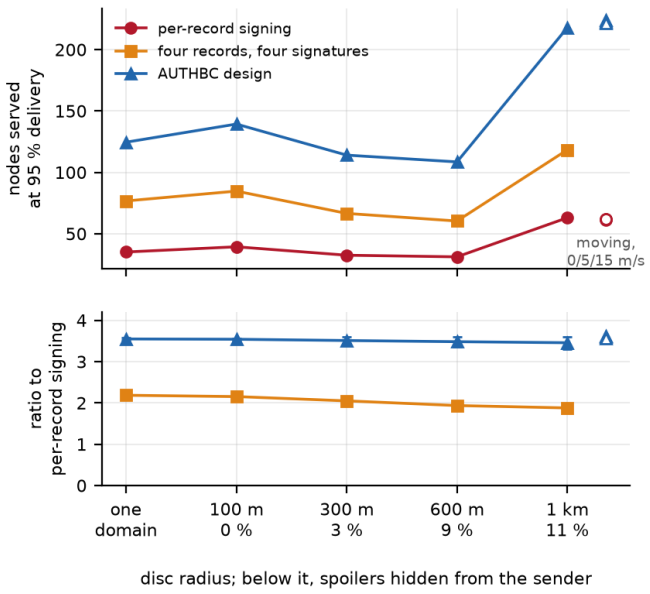


Fig. 8. Nodes served at 95% delivery as the swarm spreads (top), and the design's ratio to per-record signing (bottom); hollow markers: moving at 0, 5 and 15 m/s.

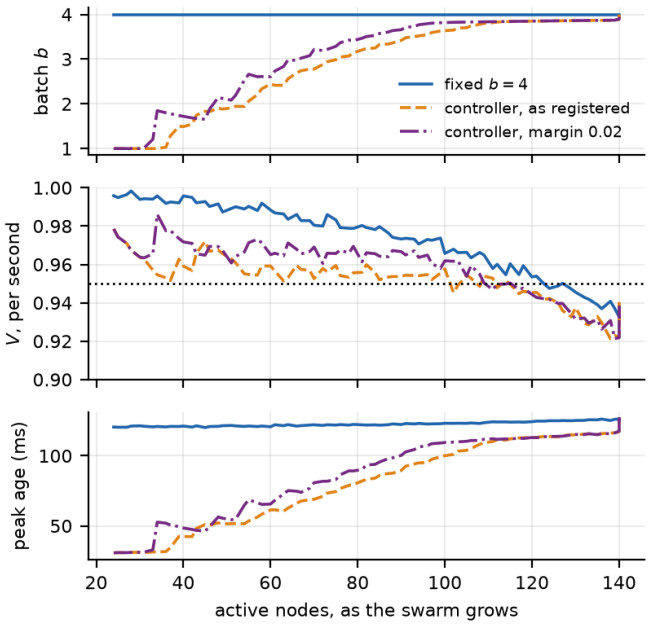


Fig. 9. A swarm growing in flight (ten seeds): the batch each policy chooses, the share of records verified in each second, and the peak age of information.

Receiver CPU: Equation (4) charges cryptography only. By it, one core of the Raspberry Pi 4 serves a neighborhood of 296 nodes when each frame carries one Ed25519 signature and of 77 when every record is signed. At its $N_{\max}$ the design uses 42% of a core, so by this count the channel binds first. Row 6 shows that the count can bind: with a signature on every record it reaches the limit of the channel at 98% of a core. With BLS a verification takes 8.61 ms and one core

TABLE IX
TIME TO RECEIVE A FOUR-RECORD FRAME ON ONE RASPBERRY PI 4 CORE, AND THE NODES ONE CORE SERVES; ALL FOUR TIMED IN ONE SESSION.

Receiver	μs per frame	Nodes per core
interpreted (Python)	1 409	57
compiled (Rust)	277	289
compiled, batches of 64	124	643
lazy: tag and decode, then batches	10.2 + 117.2	628

serves 10 nodes: on this hardware BLS is excluded by CPU well before the channel is reached, in addition to its larger signature. Ed25519 was designed for batch verification [34]: on the Pi 4 a batch of 64 takes 0.46 of the time per signature. The prototype verifies one signature at a time, and ECDSA does not admit batches.

The receiver we built does more than verify, and for it the order is reversed. Decoding a frame and rebuilding its records, in an interpreted prototype, bring the time to receive a four-record frame on the Pi 4 to 1.43 ms, 5.5 times its verification. By that time one core serves 57 nodes: the processor binds before the channel. Run as one receiver process per core, one to four at once, the board serves 56, 106, 150 and 187 nodes. Four cores give 3.36 times one, less than we had written down beforehand, so 124 nodes need three of them. Compiled in Rust, the same receive path takes 0.277 ms on the Pi 4, 5.1 times less, and the channel binds again (Table IX). A receiver that uses records once their group tag and frame are checked and verifies their signatures in batches before they enter the ledger serves 628 nodes per core. A record is then evidence within 191 ms by the convention of Eq. (2); in that window a member holding the group key could have records acted on that are later purged.

Flooding: A forged frame — a real sender, a newer sequence number and a well-formed signature by another key — costs the Python receiver 1.08 ms on the Pi 4, against 1.44 ms for an honest one, which also pays for its storage. With 41 senders on one receiver core, 247 such frames per second push honest delivery below 0.95. An 8 B group tag refuses the frame of an outsider in 22.5 μs: at 3000 forged frames per second, more than the 2 857 that one channel carries, honest delivery stays at 0.984. A member holding the key is stopped neither by the tag nor by a per-sender rate cap: honest delivery falls below 0.95 from 129 forged frames per second behind the tag, and from 123 behind the cap when the forgeries name every sender (Fig. 10). Against one forging sender the cap keeps delivery at 0.96 or more, up to the rate one channel carries. The tag costs 3.1% of capacity.

Other PHY Rates: The crossings were also simulated on 802.11a at 12 and 24 Mb/s and on the 10 MHz channel of 802.11p at 3, 6 and 12 Mb/s. The design serves 3.39–3.80 times the nodes of per-record signing, more at faster rates: a faster PHY shortens the payload but not the fixed cost of a frame (Table X). The access-rule model stays within 1.5% except at the fastest rate of each channel width, where it is

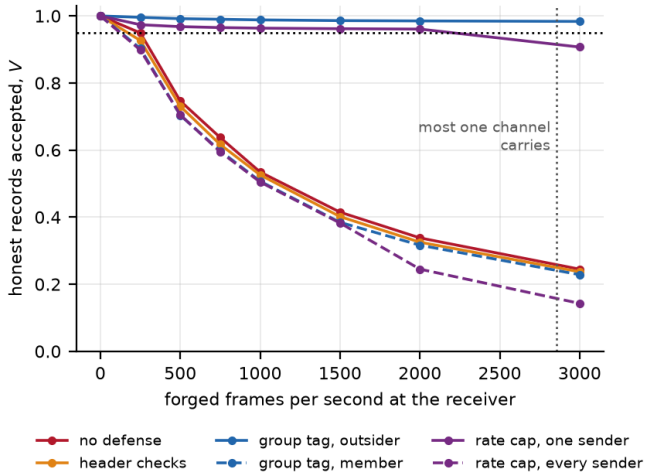


Fig. 10. Honest delivery on one Raspberry Pi 4 receiver core with 41 senders, against the rate of forged frames, for each defense and attacker.

TABLE X
NODES SERVED AT 95% DELIVERY IN NS-3 ON OTHER PHYS AND WITH A POST-QUANTUM SIGNATURE, AND THE ACCESS-RULE MODEL'S ERROR (NEGATIVE: OPTIMISTIC).

PHY and rate	Frame (μs)	$N_{\max}$ per rec.	$N_{\max}$ design	Ratio	Model (%)
802.11a, 6 Mb/s	338	35.3	124.7	3.54	+0.5
802.11a, 12 Mb/s	198	50.5	184.2	3.65	−1.5
802.11a, 24 Mb/s	126	63.0	239.7	3.80	−4.6
802.11p, 3 Mb/s	666	20.4	69.1	3.39	+0.4
802.11p, 6 Mb/s	386	30.7	109.6	3.58	−1.1
802.11p, 12 Mb/s	242	41.2	151.6	3.68	−4.0
Falcon-512, 802.11a 6 Mb/s	1 142	12.4	43.5	3.5	+0.6

Frame: the design's, on air. Model: the larger gap of the two rows.

2.9–4.6% optimistic.

A Post-Quantum Signature: Falcon-512, with its padded signature of 666 B, fits a frame. Simulated as above (Table X), per-record signing serves 12.4 nodes and the design 43.5, 3.5 times as many, as with Ed25519: the gain comes from sending a quarter as many frames, whatever the size of the signature. On the Pi 4 Falcon-512 signs in 638 μs and verifies in 88 μs, faster than Ed25519 verifies.

E. Certificates

The rows so far assume that receivers already hold the public keys of the senders. Vehicular practice sends a full certificate in every fifth message and an eight-octet digest in the others [9]. Charging that to every frame — a 162 B certificate, so 38.8 B per frame on average — gives the first "with cert." column of Table V (baseline: 184.57 B vs. design: 52.95 B per record). That reading favors the design: the practice it copies sends ten messages a second, so one in five is one certificate in 500 ms, and a baseline of 50 frames/s charged per frame sends five times that. The second column charges one certificate per 500 ms at any frame rate (baseline: 159.93 B vs. design: 51.41 B); batching then amortizes only

TABLE XI
SENDER ENERGY PER RECORD ON A RASPBERRY PI 4 (μJ): METERED (MEDIAN AND RANGE OF FIVE REPETITIONS) AGAINST TIME × POWER.

	Metered	Model	Gap
first format, every record signed	118.8 [118.5, 119.9]	105.8	12.3%
first format, one signature per 4	58.4 [58.1, 59.3]	54.3	7.5%
lean format, every record signed*	273.1 [270.6, 276.9]	284	−3.9%
lean format, one signature per 4*	155.4 [155.1, 155.8]	163	−4.4%
compiled, every record*	60.7 [60.2, 62.4]	70.2	−13.6%
compiled, one per 4*	17.1 [16.9, 17.1]	19.4	−11.6%

* Lean rows: second board and sensor.

TABLE XII
RECORDS FROM 30 FLIGHTS OF THE AUTOPILOT'S SIMULATOR, TEN PER AIRFRAME, AT 50 RECORDS/S: SIZES IN BYTES, AND WHAT THE DESIGN SAVES.

		Delta				
Fields	**Key**	**quad**	**wing**	**VTOL**	**Saved**	**Ratio**
9	22.8	9.0	9.2	9.2	70.1%	3.51
15	32.8	16.0	16.2	16.2	67.2%	3.31
25	45.5	26.0	26.3	26.2	63.3%	3.04

Saved: bytes on air. Ratio: nodes served, design over per-record signing, by the access-rule model.

the digest. These are explicit certificates; implicit ones are shorter (a signed message of at most 226 B against 330 B [9]) and are not priced, so both columns are upper bounds.

F. Energy

Table XI presents the metered sender energy in both formats, revealing three key findings:

1) *Batching Saves Energy:* In the first format, encoding, hashing and signing a record costs 58.4 μJ batched against 118.8 μJ when every record is signed, 51% less. The lean sender meters 155.4 μJ per record against 273.1 unbatched, 43% less.
2) *Model Agreement:* The model of the first format composes separately measured times with one power, the median over these same metered runs, so the comparison tests the times: the model is 7.5–12.3% low, the gap being frame assembly in an interpreted prototype. The lean sender, whose model is its timed frame (0.87 ms for four records) at that power, reads −4.4% and −3.9% against its model. Its rows were metered on a second board and sensor, where a control repeating the first row gives 116.2 μJ.
3) *Bytes Versus Energy:* In the interpreted prototype the order between the formats is the reverse of the bytes: the lean sender costs 2.7 times the energy per record of the first format: its codec, written for the size of the frame, is, in this prototype, the slower to run. Compiled in Rust, it costs 17.1 μJ per record, 3.4 times less than the first format with batching: the order follows the bytes, and the reversal was the interpreter's. Its model reads −11.6% and −13.6%, because compiled code draws 0.66 W above idle, less than the one power of the model; a control repeating the first row that day gives 118.0 μJ.

Receiver and radio energy are modeled, not metered, and for the first format only, the radio's power coming from a unicast measurement on another band: adding one receiver and the radio, the model gives 131.7 against 376.8 μJ per record.

G. Representativeness of the Synthetic Telemetry

Every size above comes from one generator, whose delta record is exactly 9.0 B in every sample — the floor of the format. Twelve public PX4 flight logs (four quadrotors, two hexa/octorotors, three fixed wings, three VTOLs; 29 092 in-flight records), chosen by a rule fixed before any was opened, were passed through the same encoder. The logs carry position at 5 Hz, so the comparison is at 0.2 s spacing and above, not at the 50 Hz of the operating point.

Flight Logs: At 0.2 s a real delta record averages 11.1 B against the 10.8 B of the generator, and a real keyframe 22.0 B against 24.0. The spread is by speed: the six multicopters, mostly hovering, average 10.0–10.5 B and the three fixed wings up to 13.2 B. With real sizes at 0.2 s the design saves 69.1% (68.2% in the least favorable log) where the generator gives 69.4%. The generator is representative to within about one percentage point, and optimistic for fast vehicles.

Operating Rate: For the operating rate itself we flew PX4 in software-in-the-loop — its own estimator, logger and MAVLink module on a simulated quadrotor — with position logged every 20 ms. Over 14 317 in-flight records, at up to 10 m/s, every delta record is 9.0 B, the floor of the format; the keyframe averages 20.9 B and the design costs 42.47 B per record, 1.8% less than with the records of the generator. The 50 Hz position stream of the same autopilot holds a 20.0 ms period to within ± 0.4 ms (central 68%).

Richer Records and Other Airframes: 30 flights of the same autopilot's simulator, ten each of a quadrotor, a fixed wing and a VTOL, the fixed wings at their planned 15–25 m/s, give records of 9, 15 and 25 fields at the 20 ms spacing of the operating point. Almost every field changes by less than its first byte holds, so a delta costs about a byte per field on every airframe, a fixed wing 0.2 B more than the quadrotor. With 15 fields, adding attitude and body rates, the design saves 67.2% of the bytes and serves 3.31 times the nodes; with 25, adding navigation quality, airspeed, wind, mission state and thrust, 63.3% and 3.04 times. The gain is set by the number of frames, not by the size of the record (Table XII).

H. Validation of the Models

Models Against the Simulator: The broadcast model [27] matches saturated ns-3 runs within $\pm 0.51\%$ over $N = 5$–50 (30 seeds).

Access Rule Against Two Radios: On two Raspberry Pi 4 radios, one transmitter alone loses 4 frames in 17 600 ($p = 2.3 \times 10^{-4}$). With both radios saturated, 12.4% of the frames put on air are lost where the access rule gives 11.8%, a prediction registered before the run. Below saturation the agreement is poorer: as the medium fills from half to 85% the radios lose 1.13, 0.79 and 0.56 times what the model

TABLE XIII
BYTES PER MESSAGE OF CERTIFICATELESS AGGREGATE SCHEMES AS REPORTED IN [7] (WITH A 67 B MESSAGE), AND PER RECORD OF AUTHBC WITH CERTIFICATES CHARGED PER FRAME.

Scheme	Bytes per Message
Wang *et al.**	859
Liang *et al.**	735
Xu *et al.**	596
Cahyadi *et al.**	583
Li *et al.* [7]	583
AUTHBC baseline (row 5), with certificates	184.57
AUTHBC design (row 8), with certificates	52.95

* total per message as reported in [7].

TABLE XIV
STREAM-SIGNING SCHEMES AT 50 RECORDS/S, SIMULATED AS IN TABLE V; IN TIME: A RECORD COUNTS ONLY IF IT VERIFIES WITHIN $D_{\max}$.

Scheme	Auth. B/rec.	Total B/rec.	N_{sat}	$N_{\max}$	In time
MAVLink 2 tag†	13	93.8	23	42 [41, 42]	42.1
TESLA†‡	24	105.8	22	40 [40, 40]	39.6–39.8§
Gennaro–Rohatgi‡	32	113.8	21	39 [39, 39]	none
EMSS‡	65	147.3	20	34 [34, 34]	none
Wong–Lam, block of 4	129	210.8	17	28 [28, 29]	29.1
signature per record	64	145.8	20	35 [35, 35]	35
AUTHBC design	16	43.3	52	124 [124, 125]	124

† no non-repudiation. ‡ a packet does not verify alone.
§ disclosure delay of two intervals of at most 20 ms.

predicts, and the registered prediction held at the first two loads and failed at the third. The cause is not established. Below saturation a radio often holds two frames of its own, and a radio of this family sends its queued frames closer together than the standard allows — one per 2.019 ms against 2.078 ms — as commercial cards are known to [48]. That state is common with two busy radios and rare at the operating point. However, a follow-up that switched off the frame bursting of the vendor, with four further predictions registered, failed all four: bursting has no demonstrated effect on the loss. No capacity was measured: the capacities in this paper rest on simulation.

IX. COMPARISON WITH PUBLISHED SCHEMES

Stream Signing: Table XIV places the schemes of Section II at our operating point. Each was implemented and run in ns-3 in the lean one-record frame, with its own authenticator in place of the signature. All of them send a frame per record — 50.0 frames/s per node (50.5 for EMSS, with its signature packets) where the design sends 12.5 — so the fixed channel time of a frame is paid for every record whatever the size of the authenticator. A 13 B tag in place of a 64 B signature raises $N_{\max}$ from 35 to 42 nodes and gives up non-repudiation; four records in a frame raise it to 124. Counting a record only when it verifies within the 100 ms deadline, only MAVLink 2, Wong–Lam and TESLA with a short disclosure delay reach 95%; Gennaro–Rohatgi and every variant of EMSS never do, because a record waits for packets sent after it. Tree chaining is the one scheme that keeps what a ledger requires — every packet verifies alone and its origin can be shown to a third party — and it pays 129 B per packet for it. These schemes amortize the cost of signing; on a contended channel the cost that binds is the frame.

Aggregate Signatures: The nearest published comparators are certificateless aggregate signature schemes for vehicular broadcast. Table XIII lists what one signer sends per message. For n messages that is n times as much, since each carries its own identity, key and signature components. Aggregation acts afterwards, at a roadside unit: it reduces the work of the verifier and what the unit forwards — in the one scheme read at source, 388 B for a single signature and 784 B for the aggregate of a hundred [49]. Batching acts before, on the own records of a sender. The two act on different links, and the table should not be read as a ranking: those schemes provide conditional privacy, which AUTHBC does not; their group elements are 128 B where our signature is 64 B; and their values are for a different message. What the comparison does show is that per-message authentication of this kind costs several hundred bytes, and that aggregation alone does not change that.

TABLE XV
CONFIGURATION SELECTION DECISION MATRIX.

Scenario	Signing	b	Binds	Outcome
802.11, $\Lambda D_{\max} = 5$	per frame	4	freshness	43.25 B/record†
802.11, $\Lambda D_{\max} \leq 2$	per record	1	freshness	no batch admitted
LoRaWAN, DR4–DR7	per frame	8*	duty cycle	0.20 records/s
LoRaWAN, eight rates	none	–	payload	no frame fits§
Interpreted receiver	per frame	4	CPU	57 nodes per core
Compiled receiver	per frame	4	channel	289 nodes per core
BLS12-381	not advised	–	CPU	10 nodes per core

† $N_{\max} = 124$ in simulation; no capacity was measured.
* at DR5 with a 242 B frame under the 1% duty cycle.
§ under the five scope conditions of Section VII.

X. DISCUSSION AND DEPLOYMENT GUIDANCE

A. Configuration Selection Decision Matrix

Table XV maps each deployment scenario to the configuration that the constraints of Section V admit, and to the constraint that binds.

B. Interpretation of Key Findings

The 3.5× capacity gain of the lean design stems from the fixed channel cost of a frame and only secondarily from its payload: a frame costs about 173 μs whatever its length, so a node that sends 12.5 frames/s where per-record signing sends 50.0 removes three of every four such costs. Sharing the frame alone therefore yields 2.2×, and sharing the signature a further 1.6×.

The batch of four is a consequence of the freshness bound and not of the MTU: the fill time b/Λ dominates Eq. (2), whereas the MTU would admit 151 records. The keyframe in every frame is a consequence of the verifiability target: with

TABLE XVI
RECORDS PER FRAME EACH BROADCAST SYSTEM ADMITS WITH ONE SIGNATURE PER FRAME, BY DEADLINE, UNDER EQ. (2); —: NO SIGNATURE FITS.

System	Payload (B)	Rate (/s)	$D_{\max}$ 100 ms	200 ms	300 ms	1 s
ADS-B 1090ES	7	2	—	—	—	—
Remote ID, Bluetooth 4	25	1	—	—	—	—
Remote ID, Bluetooth 5	225	1	none	none	none	none
V2X CAM, 10 Hz	1500	10	none	1	2	9
V2X CAM, 1 Hz	1500	1	none	none	none	none
this design, 802.11a	1500	50	4	9	14	49

a target equal to the loss rate, Eq. (3) admits $r = 1$ only, and the most a longer interval could save is 3.5 B per record.

The neutrality of the signature scheme reflects that Ed25519 and ECDSA produce frames of one size and differ in verification time only, which enters the CPU constraint and not the bytes. The exclusion on LoRaWAN has two distinct causes: five data rates cannot hold a 64 B signature in any format, and three cannot hold the header of this format beside a signature and a chain link.

The same accounting applies to other broadcast systems. ADS-B 1090ES and Bluetooth 4 Remote ID cannot carry a signature at all. Bluetooth 5 and Wi-Fi Remote ID fit up to seven records in a frame, but at one message a second no frame meets any deadline up to 1 s. V2X CAM at 10 Hz admits no frame at 100 ms, one record at 200 ms and two at 300 ms (Table XVI). This is consistent with these standards signing each message; it does not explain why they do.

C. Deployment Recommendations

Our findings enable the following configuration guidance; the capacities behind it are simulated. (1) *High-rate 802.11 swarms* ($\Lambda D_{\max} = 5$): one Ed25519 signature and one chain link per four records with a keyframe in every frame, at 43.25 B per record (per-record signing: 145.77 B). (2) *Tight deadlines* ($\Lambda D_{\max} \leq 2$): per-record signing, since no batch meets the deadline. (3) *LoRaWAN provenance links*: the four feasible data rates only, with the batch sized at the record spacing it implies; on the other eight rates a signed, chained frame that verifies alone cannot be carried, and Table IV lists what each relaxation buys. (4) *Constrained receivers*: the processor is provisioned before the channel, since the interpreted receiver serves 57 nodes per core, a compiled one 289, and BLS 10. (5) *Loss budget*: ϵ set at least 0.02 above the expected frame loss, since at $\epsilon = p$ the design meets its target with no margin (Fig. 4). (6) *Outsiders*: an 8 B group tag, at 3.1% of capacity. These recommendations are summarized in Table XV.

XI. LIMITATIONS

Capacities Are Simulated: On hardware, contention was measured between two radios only (Sec. VIII-H). The capacities in Table V come from ns-3 with one collision domain, equal received power and static nodes. Unequal power with capture was simulated separately and raises them (Sec. VIII-D); a swarm spread over a wider area, with hidden stations, and a moving swarm were simulated as well, with no Doppler and no time-correlated fading. Send instants are redrawn within each period (Sec. VI). The own stream of PX4, in simulation, holds its period to about ± 0.4 ms, which is nearer to the strictly periodic source and its wider spread between runs.

Telemetry Is Synthetic: Record sizes come from a generator. The check against flight logs covers twelve logs of one autopilot at 5 Hz position resolution, and simulated flights of that autopilot at 50 Hz, whose navigation-quality fields do not vary.

Loss Models: The loss experiment uses independent, Gilbert and length-dependent loss generated at the frame level, and collision loss traced in ns-3; no loss was measured on air. Under length-dependent loss the design is below its target where the frame of the baseline loses 5% (Sec. VIII-B). Batches of two, three and eight were simulated for the capacity rule alone, and nothing else here is evaluated at them.

Energy and CPU: Only the sender is metered, with a sensor never checked against a reference load; receiver and radio energy are modeled. The CPU share of the receiver counts verification and hashing, not decoding; counting both, the interpreted prototype needs three of the four cores of the board for the neighborhood of the design. The compiled receiver was timed alone, without a radio or a dispatcher in the path.

Security Scope: Confidentiality, availability and key management are outside the design (Sec. III). A swarm member holding the group key can still flood, and a delayed frame is refused only while GNSS time is held. The proofs use three models, not one, and assume canonical encoding injective. With one link per frame, continuity inside a frame rests on the frame signature alone, and showing one record to a third party means showing its frame. The receiver therefore keeps every accepted frame, at its on-air length for every b records held.

Post-Quantum Signatures: Falcon-512 fits a frame (Sec. VIII-D). An ML-DSA signature is 2420 B [9] and exceeds the 1500 B MTU with any header, so it cannot ride in one frame at any batch size; freshness caps the batch at four, so it could not be amortized either. Such a signature would need a design built around fragmentation [9], [50].

Multi-Hop: Relayed records are not evaluated. A relay cannot re-sign: it carries the signatures of the originators or one BLS aggregate [14]. Both are implemented; neither is measured here.

XII. CONCLUSION

Authenticated UAV telemetry on a hash-chained ledger imposes a cost that we did not find accounted for jointly in prior work: a signature and a chain link larger than the record they protect, on a channel where bytes are airtime. This work addresses that gap by costing authentication at the frame and not at the record, through emitted frames, ns-3 simulation with

30 seeds per point, and metered sender energy; what a frame must carry decides where it can be sent.

Two feasibility boundaries result. A frame that verifies alone needs a header, a chain link, a signature and one self-contained record: eight of the twelve EU863-870 LoRaWAN data rates cannot carry that at any batch size, five because the signature alone does not fit and three because of the own header of this frame. Where the frame fits, one signature and one chain link per four records cut the bytes per record by 70.3% (145.77 B vs. 43.25 B) and let one 802.11a collision domain carry 124 nodes where per-record signing carries 35. Two constraints shape the design: the batch is capped by freshness, not by the MTU, and a frame that depends on its predecessor does not meet a verifiability target equal to the loss rate. Batching and header design decide what can run; among 64 B signatures the scheme does not. The capacity values are simulated, also for a spread and moving swarm, where the gain stays 3.46–3.63-fold; on hardware the access rule behind them held between two saturated radios and not at every load below saturation, and no capacity was measured. Compiled, the receiver serves 289 nodes on one core of a Raspberry Pi 4, and the lean sender spends 17.1 μJ per record, a third of what the first format spends.

This accounting — what a frame that verifies alone must carry, and which constraint binds — is not specific to FANETs and should carry over to other broadcast systems whose records are small beside their authenticator; this paper evaluates it for UAV telemetry only. Future work should address the 5G NR sidelink, the 3GPP link for direct broadcast between vehicles, and a variant for post-quantum signatures that do not fit one frame.

Appendix A
AUTHBC Frame Layout

For reproducibility, the complete layout of the lean frame is provided. A frame is one canonical CBOR map with the integer keys 0–7:

```
frame = {
  0: 2,           ; v: format version
  1: 0 / 1 / 2,   ; t: frame type
  2: uint,        ; src: sender id (16 bit)
  3: uint,        ; base_seq (32 bit)
  4: 1..255,      ; n: records in frame
  5: bytes(32),   ; link: prev_hash of rec 0
  6: bytes,       ; recs: record stream
  7: bytes(64)    ; auth: one signature, or
     / [n * bytes(64)]     ; one per record
}
rec = 9 varints: ts lat lon alt
      vel_x vel_y vel_z battery mode
```

Key Implementation Notes:

- *Frame Types:* Type 0 carries one signature per record; type 1 carries one signature and decodes alone; type 2 carries one signature and codes its first record against the last record of the previous frame.
- *Record Coding:* A keyframe record carries absolute values, with `ts`, `battery` and `mode` unsigned and the other fields zig-zag signed; a delta record carries zig-zag differences from the record before it. All values are LEB128 variable-length integers.
- *Derived Fields:* The sender identifier and sequence number of record i are `src` and `base_seq` $+ i$. Every `prev_hash` after the first is recomputed by the receiver and never read from the frame.
- *Signed Bytes:* A signature covers the canonical CBOR encoding of the rebuilt ledger records, not the record stream.

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